AIDAN SIMS

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EDUCATION

CORNELL UNIVERSITY

M. Eng. in Engineering Physics, GPA: 3.9, December 2024

- **Relevant Coursework:** Nanomaterials: Chemistry and Physics, Mathematical Physics, Cryptography, Solid State Physics, Computational Methods in Physics

B.A. in Computer Science, Minor in Business, **GPA: 3.7 (Dean's List)**, May 2024

- **Relevant Coursework:** Quantum Computing, Quantum Physics and Engineering, Quantum Information Processing, Artificial Intelligence, Analysis of Algorithms, Risk Simulation and Monte Carlo Methods

RESEARCH PROJECTS AND GRANTS

- **Open Tavis-Cummings Model Simulation (**https://arxiv.org/abs/2501.18522 **)**: Implemented quantum algorithms from abstract algorithm descriptions in literature to efficiently simulate TC model behavior.
- **PyCVXQuad**: Awarded Unitary Fund grant to implement a Python library for semidefinite matrix logarithm approximation. Expanding library to cover quantum information functions with novel approximations.

EXPERIENCE

SCALE AI

Forward Deployed Engineer *(Jun 2025 – Present)*

- Develop software to improve data delivery process to large AI labs.

CORNELL UNIVERSITY, Ithaca, NY

Research Assistant, Mark Wilde Lab *(Sep 2023 – Feb 2025)*

- Led two research projects in quantum algorithms and quantum information. Presented recent updates in quantum computing literature through group meeting presentations.

Foundations of Artificial Intelligence (CS 4700/5700), Teaching Assistant *(Dec 2021 – Jun 2023)*

- Created assignments and slides. Reinforced students' understanding of material through office hours.

CAPITAL ONE

Commercial Banking Technology, Software Engineering Intern *(Jun 2023 – Aug 2023)*

- Developed underlying algorithms for a Python tool that compares treasury cash management (BAI2) files.
- Implemented front-end features for a customer facing webpage.

SURGE AI

Code Generation and Search Evaluation teams, Artificial Intelligence Intern *(Apr 2022 – Present)*

- Train large language models on code and natural language generation tasks so that they are more helpful and aligned (RLHF). Perform consumer research on search algorithm ranking preferences.

ASTRAZENECA

Rare Disease Business Unit, Software Robotics Intern *(Jun 2021 – Aug 2021)*

- Assessed benefits and drawbacks of various business automation software ahead of business-wide rollout.
- Implemented process which tracks rare disease drug shipments, reducing costs by ~$300k annually.

TECHNICAL SKILLS

Qiskit, Pennylane, Python, Java, OCaml, C, AI/ML, SQL, HTML, CSS, JavaScript, Word, Excel, PowerPoint, RPA Software (UIPath, Blue Prism), Google Suite, Git, AWS